

CONFIDENTIAL — INTERNAL USE ONLY

CBSE Class 10 Question-Type Research & AI Prompt Engineering Report

AI-Assisted Content Creator Internship — Task Submission

Prepared for: samaranai.in

Ownership & confidentiality notice

All content in this document — including the research, question-type analysis, AI prompts, generated questions, answer keys and explanations — is proprietary work product created for and owned by samaranai.in.

This document and its contents must not be posted, published, shared, or reproduced in any public forum, repository, portfolio, or social platform, in line with the internship's confidentiality instruction.

0. Problem Statement

This report was produced as part of an AI-assisted content creator internship task assigned by samaranai.in. The brief, reproduced below, is met in full within this single document.

Task 1 — Study CBSE Class 10 board question types and expected standards, and prepare a subject-wise report (e.g. assertion-reasoning, conceptual, case-based questions — types differ by subject).

Task 2 — Prepare AI prompts to generate these question types for one chapter of each subject.

Task 3 — Create sample MCQs for Science and Mathematics, one chapter each, as a proof of concept.

Deliverable — A single Word document / PDF, submitted within 24 hours, containing the problem statement, an ownership note (content belongs to samaranai.in), the full solution, the steps followed, and the final output.

1. Subject-wise CBSE Class 10 Question-Type Report

CBSE's Class 10 board exam for the 2025–26 academic session (first phase held February–March 2026) follows a competency-based design aligned with NEP 2020. Roughly 50% of every theory paper is competency-based (case-based, source-based, data-interpretation and situational questions), and question papers are offline, split into five broad sections (A–E) that combine objective and descriptive formats. Each subject carries 80 marks for theory and 20 for internal assessment (100 total), with no negative marking anywhere.

Sources consulted: CBSE sample papers and marking schemes for 2025–26 (Mathematics Standard/Basic, Science, Social Science), and secondary compilations from Learn CBSE, Shaalaa, Vidyaarohi, PW Live and Shiksha, cross-checked against the board's own five-section (A–E) design.

1.1 Mathematics (Standard & Basic) — verified pattern

Mathematics follows the clearest, most stable five-section structure, confirmed directly from the released 2025-26 sample paper and marking scheme:

Section	Q. Nos.	Question type	Marks each	Total
A	1–18	Multiple Choice Questions (MCQ) — pure recall/application, single correct option	1	18
A	19–20	Assertion (A) — Reason (R) — judge truth of both statements and the logical link between them	1	2
B	21–25	Very Short Answer (VSA) — 2-step conceptual/computational questions	2	10
C	26–31	Short Answer (SA) — multi-step problems, proofs, constructions	3	18
D	32–35	Long Answer (LA) — full derivations, word problems, geometry proofs	5	20
E	36–38	Case-study / source-based — a real-world scenario with 3 sub-parts (1+1+2 marks)	4	12

Internal choice is offered in select questions across Sections B–E (not in Section A). No overall subject choice exists. Standard-level papers lean more on algebraic manipulation and proof; Basic-level papers emphasise direct application and numerical fluency of the same topics.

Expected standard / cognitive demand

- MCQs and A–R: test conceptual clarity and the ability to spot common misconceptions (e.g. "product of two irrationals is always rational" — false).
- VSA/SA: expect a clean, labelled step-by-step method — marks are awarded per correct step, not only for the final answer.
- LA and case-based: expect real-world framing (EMI, height-and-distance, sports data) that must be translated into an equation/model before solving.

1.2 Science — Physics, Chemistry, Biology

From 2025-26 onward, the Science paper is explicitly divided into three content sections — Biology, Chemistry and Physics — inside the same 80-mark, five-format (A–E) design, with 39 compulsory questions overall. This is a structural change from earlier years, where the three branches were interleaved.

Section	Format	What it tests	Typical weight
A (across all 3 branches)	MCQ + Assertion–Reason	Direct recall, identification of reaction/process type, reading graphs/diagrams	~20 marks
B	Very Short Answer (1–2 mark)	One-line definitions, balancing equations, naming/labelling	~10 marks
C	Short Answer (3 mark)	"Why/how" conceptual reasoning, short numericals (lens/mirror, electricity)	~18 marks
D	Long Answer (5 mark)	Multi-part numericals, diagram-based explanations, life-process pathways	~20 marks
E	Case-based (4 mark)	A passage/experiment/graph followed by 3 linked sub-questions	~12 marks

Expected standard / cognitive demand

- Heavy use of Assertion–Reason on cause-effect science facts (e.g. rusting is oxidation vs. rusting needs both O₂ and moisture).
- Case-based questions typically embed a lab observation, a real-life application (e.g. food preservation, eye defects, electricity bill) or a graph, testing application over memorisation.
- High-weightage chapters for 2025-26 per released blueprints: Chemical Reactions & Equations, Acids/Bases/Salts, Carbon and its Compounds, Life Processes, Light — Reflection & Refraction, Electricity.

1.3 Social Science — History, Geography, Political Science, Economics

Social Science is also restructured into four content sections for 2025-26: Section A – History, Section B – Geography, Section C – Political Science, Section D – Economics — each within the same MCQ/VSA/SA/LA/Case-based spread, and answers must be written under the matching section or they are not evaluated.

Question type	Description	Notes
MCQ	Single-line factual or map-based recall	Includes CAP-based (map-pointing) questions in Geography
Assertion–Reason	Statement + reason pair on a historical/economic cause-effect link	Common in History and Economics
Source-based (VSA/SA)	A short excerpt/cartoon/photograph followed by 2–3 questions	Central to the "competency-based" 50% weightage
Map-based	Locate/identify/label on an outline map (mainly Geography, sometimes History)	Marked separately, no negative marking
Long Answer	Explain causes/effects/processes in structured paragraphs	Rewards use of correct terms, examples, sequencing
Case-based	An unseen passage or data table (e.g. population pyramid, budget data) with linked sub-questions	Tests application, not recall

1.4 English (Language & Literature)

English does not follow the same A–E numeric layout as the science/commerce group; it is organised by skill area instead:

Section	Question type	What it tests
Reading	Unseen passage(s) (factual + discursive/literary) — mostly MCQ-based sub-questions, some VSA	Inference, vocabulary-in-context, tone, main idea
Writing & Grammar	Formal/informal writing (letter, notice, analytical paragraph) + editing/omission/transformation exercises	Format, coherence, register, grammatical accuracy
Literature (First Flight / Footprints)	Extract-based MCQs (poetry/prose) + short/long reference-to-context and character/theme questions	Comprehension, textual evidence, personal response

1.5 Hindi (Course A / B)

Section	Question type	What it tests
अज्ञात पाठ्य/कविता (Unseen passage/poem)	MCQ + short answers	Comprehension, inference, vocabulary
व्याकरण (Grammar)	MCQ / fill-in / transformation (अभिप्रेत, अर्थपूर्ण, अर्थ-अभिप्रेत, अर्थपूर्ण)	Rule application, not just definition
अभिप्रेत/अर्थपूर्ण — अर्थपूर्ण अर्थपूर्ण (Kshitij/Sparsh)	Extract-based MCQs + अर्थपूर्ण/अर्थपूर्ण अर्थपूर्ण (short/long answer) on theme, character, poetic device	Textual understanding, appreciation
लेखन (Writing)	अभिप्रेत/अर्थपूर्ण/अर्थपूर्ण (letter, paragraph, essay)	Structure, expression, originality

2. AI Prompts — One Chapter per Subject

Below are ready-to-use prompt templates, one set per subject, each targeted at a single NCERT chapter and mapped to that subject's CBSE question types from Section 1. Each prompt specifies output format, difficulty spread and the requirement for an answer key — all needed for the content pipeline to produce exam-ready material without manual re-formatting.

2.1 Science — Chapter: "Chemical Reactions and Equations"

System framing to prefix every prompt below:

"You are a CBSE Class 10 Science question setter following the 2025-26 competency-based exam pattern. Base all questions strictly on the NCERT chapter 'Chemical Reactions and Equations'. Do not introduce content outside the NCERT syllabus."

1. MCQ: "Generate 10 multiple-choice questions (1 mark each) from 'Chemical Reactions and Equations', covering: balancing equations, types of reactions (combination, decomposition, displacement, double displacement), oxidation-reduction, and everyday effects (corrosion, rancidity). Provide 4 options per question, mark the correct option, and give a 1-line explanation for each."
2. Assertion–Reason: "Generate 5 Assertion-Reason questions from the same chapter. Each must give an Assertion and a Reason, followed by the 4 standard CBSE options (both true & R explains A / both true & R does not explain A / A true R false / A false R true). Indicate the correct option and justify in 2 lines."
3. VSA (2 marks): "Generate 5 very-short-answer questions (2 marks each) that ask students to balance a given chemical equation OR identify the type of reaction from a described observation. Include the model answer."
4. SA (3 marks): "Generate 4 short-answer questions (3 marks each) requiring a 'why/how' explanation — e.g. why respiration is exothermic, why silver chloride turns grey in sunlight — with a point-wise model answer."
5. Case-based (4 marks): "Write one case-based question set: a 60–80 word real-life scenario (e.g. a rusted iron gate, or milk turning sour) followed by 3 linked sub-questions worth 1, 1 and 2 marks respectively, testing application of reaction types and prevention methods."
6. Long Answer (5 marks): "Generate 2 long-answer questions that combine a numerical/diagram-based task (e.g. writing and balancing 3 equations of different types) with a conceptual explanation, and provide a fully worked model answer."

2.2 Mathematics — Chapter: "Quadratic Equations"

"You are a CBSE Class 10 Mathematics (Standard) question setter following the 2025-26 pattern. Base all questions strictly on the NCERT chapter 'Quadratic Equations'. Show the correct standard form $ax^2+bx+c=0$ and use the discriminant/quadratic formula correctly."

7. MCQ: "Generate 10 MCQs (1 mark each) covering: identifying a quadratic equation, nature of roots via discriminant, sum/product of roots, and solving by factorisation. 4 options each, correct option marked, 1-line reasoning."
8. Assertion–Reason: "Generate 5 Assertion-Reason questions on the relationship between the discriminant (b^2-4ac) and the nature of roots (real & equal / real & unequal / no real roots). Use the standard 4-option format and justify the answer."

9. VSA (2 marks): "Generate 5 VSA questions asking students to check whether a given equation is quadratic, or to find the discriminant/nature of roots for a given equation, with worked solutions."
10. SA (3 marks): "Generate 4 SA questions requiring solving a quadratic equation by factorisation or the quadratic formula, including at least one word problem (age, speed-distance-time, or area) translated into a quadratic equation."
11. Case-based (4 marks): "Write one case-based question set with a real-world scenario (e.g. a rectangular park's dimensions, a ball's trajectory, or a business profit model) that leads to a quadratic equation, followed by 3 sub-questions (1+1+2 marks) of increasing difficulty, with model answers."
12. Long Answer (5 marks): "Generate 2 long-answer word problems (e.g. two-digit number problems, work-and-time, or speed problems) that must be modelled as a quadratic equation and solved fully, with every step shown."

2.3 Social Science — Chapter: "Nationalism in India" (History)

13. MCQ / map-based: "Generate 8 MCQs from 'Nationalism in India' covering the Non-Cooperation and Civil Disobedience movements, Rowlatt Act, Jallianwala Bagh, and Salt March, with 4 options and correct answer marked."
14. Assertion–Reason: "Generate 4 Assertion-Reason questions linking a cause (e.g. Rowlatt Act) with its effect (nationwide protest), using the standard 4-option format."
15. Source-based (SA): "Provide a short (60-word) extract paraphrased from a period source describing peasant participation in the movement, followed by 3 questions testing comprehension and inference."
16. Case-based: "Write a case-based set built around a description of the Salt March's route and impact, with 3 linked sub-questions (1+1+2 marks)."
17. Long Answer: "Generate 2 long-answer questions asking students to explain how different social groups (peasants, workers, business class, women) interpreted 'Swaraj' differently, expecting a structured, example-based answer."

2.4 English — Chapter: "Nelson Mandela: Long Walk to Freedom" (First Flight)

18. Reading-style MCQ (extract-based): "Generate 8 MCQs based on short (unquoted, paraphrased) extracts from the chapter, testing inference, vocabulary-in-context and the speaker's tone."
19. VSA (reference-to-context): "Generate 5 short-answer questions asking students to explain, in their own words, the significance of a described moment from the chapter (e.g. the inauguration day, the idea of 'twin obligations')."
20. Long Answer / value-based: "Generate 2 long-answer questions connecting the chapter's theme of freedom and courage to a real-life or personal-response scenario, expecting a 100–120 word structured answer."
21. Writing-skill link: "Generate one formal-letter or speech-writing prompt inspired by the chapter's theme (e.g. a speech on the importance of freedom), with an evaluation rubric for content, format and language."

2.5 Hindi — □□□: "□□□□□□ □□ □□□□□" (□□□□□□□)

22. □□□□□□□□□□ □□□□□ (MCQ): "□□□□□□ □□ □□□□□" □□□ □□ □□□□□□ 8 □□□□□□□□□□ □□□□□□ □□□□□, □□□□□□ □□□□□□□□, □□□□□□ □□ □□□ □□ □□□ □□□ □□ □□□ □□□ 4 □□□□□□ □□□ □□ □□□ □□□□□□ □□□□□□ □□□□□"

- Page 8 of 14

3. Sample MCQ Bank

As requested in Task 3, full 15-question MCQ sets (with answer key and explanation for every item) were generated for one chapter each in Science and Mathematics, using the prompt templates above.

3.1 Science — Chemical Reactions and Equations (15 MCQs, 1 mark each)

Q1. Which of the following is a physical change occurring during a chemical reaction, not the reaction itself?

- (A) Change of colour
- (B) Change of state**
- (C) Formation of a gas
- (D) Formation of a precipitate

Answer: (B) Change of state — A change of state (e.g. solid to liquid) can occur without a chemical reaction; the other three are evidences of one.

Q2. $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$ is an example of a:

- (A) Combination reaction
- (B) Displacement reaction**
- (C) Double displacement reaction
- (D) Decomposition reaction

Answer: (B) Displacement reaction — A more reactive metal (Zn) displaces a less reactive one (H) from its compound.

Q3. On adding a solution of lead nitrate to a solution of potassium iodide, a yellow precipitate is formed. This reaction is:

- (A) Combination
- (B) Decomposition
- (C) Double displacement**
- (D) Displacement

Answer: (C) Double displacement — Both ions are exchanged between the two reactants, forming an insoluble PbI_2 — a double displacement (precipitation) reaction.

Q4. The chemical formula obtained after balancing: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$ is:

- (A) $\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$
- (B) $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$**
- (C) $3\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$
- (D) $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$

Answer: (B) $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$ — Balancing Fe, O and H atoms on both sides requires 3Fe and 4H₂O to give $\text{Fe}_3\text{O}_4 + 4\text{H}_2$.

Q5. Which gas is evolved when dilute HCl reacts with sodium carbonate?

- (A) Hydrogen
- (B) Oxygen
- (C) Carbon dioxide**
- (D) Chlorine

Answer: (C) Carbon dioxide — $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$; CO_2 turns lime water milky.

Q6. Rancidity of food (bad smell/taste of oily food on prolonged storage) is caused mainly by:

- (A) Reduction
- (B) Oxidation**
- (C) Decomposition
- (D) Displacement

Answer: (B) Oxidation — Aerial oxidation of fats and oils causes rancidity; antioxidants or airtight/vacuum packing help prevent it.

Q7. Which of these is an endothermic reaction?

- (A) Burning of natural gas
- (B) Respiration
- (C) Decomposition of calcium carbonate on heating**
- (D) Reaction of quick lime with water

Answer: (C) Decomposition of calcium carbonate on heating — $\text{CaCO}_3 \xrightarrow{\text{heat}} \text{CaO} + \text{CO}_2$ absorbs heat continuously; the other three release heat.

Q8. A shiny brown-coloured element X on heating in air turns black. X and the black compound are:

- (A) Fe, FeO
- (B) Cu, CuO**
- (C) Zn, ZnO
- (D) Al, Al_2O_3

Answer: (B) Cu, CuO — Copper (shiny brown) forms black copper(II) oxide, $2\text{Cu} + \text{O}_2 \xrightarrow{\text{heat}} 2\text{CuO}$.

Q9. Which of the following is used as a reducing agent to obtain manganese from manganese dioxide in a thermit-type displacement reaction?

- (A) Carbon
- (B) Aluminium**
- (C) Zinc
- (D) Sodium

Answer: (B) Aluminium — Aluminium is a strong reducing agent used in such highly exothermic displacement reactions, e.g. $3\text{MnO}_2 + 4\text{Al} \rightarrow 3\text{Mn} + 2\text{Al}_2\text{O}_3$.

Q10. Silver chloride turns grey on exposure to sunlight. This is an example of:

- (A) Thermal decomposition
- (B) Electrolytic decomposition
- (C) Photolytic decomposition**
- (D) Combination reaction

Answer: (C) Photolytic decomposition — Light energy decomposes AgCl into Ag and Cl_2 — used in black-and-white photography.

Q11. In the reaction $\text{CuSO}_4(\text{aq}) + \text{Fe}(\text{s}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$, the colour change observed is:

- (A) Colourless to blue
- (B) Blue to green**
- (C) Green to blue
- (D) Blue to colourless

Answer: (B) Blue to green — Blue CuSO_4 solution turns pale green as FeSO_4 forms; Cu is deposited on the iron nail.

Q12. Assertion (A): Corrosion of iron is commonly called rusting. Reason (R): Rusting occurs only in the presence of oxygen, without any need for moisture.

- (A) Both A and R are true, and R is the correct explanation of A
- (B) Both A and R are true, but R is not the correct explanation of A
- (C) A is true, R is false**
- (D) A is false, R is true

Answer: (C) A is true, R is false — A is correct, but rusting needs BOTH oxygen and moisture — so R is false.

Q13. Which type of reaction is represented by: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2$?

- (A) Decomposition
- (B) Displacement
- (C) Combination**
- (D) Double displacement

Answer: (C) Combination — Two reactants (CaO and H_2O) combine to form a single product, $\text{Ca}(\text{OH})_2$ — a combination reaction, and also exothermic.

Q14. Which of these is NOT a correct way to balance the skeletal equation $\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$?

- (A) $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$
- (B) $2\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$ (unbalanced, left as is)**
- (C) Balance Al atoms first, then O atoms
- (D) Use the smallest whole-number coefficients possible

Answer: (B) $2\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$ (unbalanced, left as is) — Leaving an equation unbalanced never satisfies the law of conservation of mass; option (A) is the correctly balanced form.

Q15. A double displacement reaction that also happens to be a precipitation reaction is one where:

- (A) Only gas is released
- (B) An insoluble solid is formed from two soluble reactants**
- (C) Only heat is released with no new substance
- (D) A single reactant breaks into two products

Answer: (B) An insoluble solid is formed from two soluble reactants — Precipitation reactions are a sub-type of double displacement in which an insoluble product (precipitate) separates out, e.g. $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl}\downarrow + \text{NaNO}_3$.

3.2 Mathematics — Quadratic Equations (15 MCQs, 1 mark each)

Q1. Which of the following is a quadratic equation?

- (A) $x^2 + 2x^{-1} = 5$
- (B) $\sqrt{2}x^2 + 3x - 1 = 0$**
- (C) $x^3 - x^2 + 2 = 0$
- (D) $(x-1)(x+1) = x^2$

Answer: (B) $\sqrt{2}x^2 + 3x - 1 = 0$ — It is in the form $ax^2 + bx + c = 0$ with $a \neq 0$; option (D) simplifies to $-1 = 0$, which is not an equation in x at all.

Q2. The discriminant of $2x^2 - 4x + 3 = 0$ is:

- (A) -8**
- (B) 8
- (C) 40
- (D) -40

Answer: (A) -8 — $D = b^2 - 4ac = (-4)^2 - 4(2)(3) = 16 - 24 = -8$.

Q3. If the discriminant of a quadratic equation is 0, its roots are:

- (A) Real and unequal
- (B) Real and equal**
- (C) Not real (imaginary)
- (D) Cannot be determined

Answer: (B) Real and equal — $D = 0$ always gives two equal real roots, both equal to $-b/2a$.

Q4. The roots of $x^2 - 7x + 12 = 0$ are:

- (A) 3, 4**
- (B) 2, 6
- (C) 1, 12
- (D) -3, -4

Answer: (A) 3, 4 — Factorising: $x^2 - 7x + 12 = (x-3)(x-4) = 0$, so $x = 3$ or $x = 4$.

Q5. For the quadratic equation $3x^2 + kx + 3 = 0$ to have equal roots, the value of k must satisfy:

- (A) $k^2 = 36$**
- (B) $k^2 = 9$
- (C) $k^2 = 3$
- (D) $k^2 = 27$

Answer: (A) $k^2 = 36$ — $D = 0 \Rightarrow k^2 - 4(3)(3) = 0 \Rightarrow k^2 = 36$.

Q6. The sum and product of the roots of $2x^2 - 5x + 3 = 0$ are respectively:

(A) $5/2, 3/2$

(B) $-5/2, 3/2$

(C) $5/2, -3/2$

(D) $2/5, 2/3$

Answer: (A) $5/2, 3/2$ — Sum = $-b/a = 5/2$; Product = $c/a = 3/2$.

Q7. Which value of x satisfies $x^2 - 9 = 0$?

(A) $x = 3$ only

(B) $x = -3$ only

(C) $x = 3$ or $x = -3$

(D) No real solution

Answer: (C) $x = 3$ or $x = -3$ — $x^2 = 9 \Rightarrow x = \pm 3$, i.e. two real roots.

Q8. A quadratic equation whose roots are 4 and -3 is:

(A) $x^2 - x - 12 = 0$

(B) $x^2 + x - 12 = 0$

(C) $x^2 - 7x + 12 = 0$

(D) $x^2 + 7x - 12 = 0$

Answer: (A) $x^2 - x - 12 = 0$ — Sum of roots = 1, product = $-12 \Rightarrow x^2 - (\text{sum})x + \text{product} = x^2 - x - 12 = 0$.

Q9. Assertion (A): The equation $x^2 + 4x + 5 = 0$ has no real roots. Reason (R): A quadratic equation has no real roots when its discriminant is negative.

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R does not explain A

(C) A is true, R is false

(D) A is false, R is true

Answer: (A) Both A and R are true, and R is the correct explanation of A — $D = 16 - 20 = -4 < 0$, confirming no real roots, and R correctly states the general rule that explains A.

Q10. The value(s) of k for which $x^2 + kx + 4 = 0$ has real and distinct roots is:

(A) $k > 4$ or $k < -4$

(B) $-4 < k < 4$

(C) $k = \pm 4$

(D) $k = 4$ only

Answer: (A) $k > 4$ or $k < -4$ — $D > 0 \Rightarrow k^2 - 16 > 0 \Rightarrow k^2 > 16 \Rightarrow k > 4$ or $k < -4$.

Q11. Using the quadratic formula, the roots of $x^2 - 2x - 3 = 0$ are:

(A) 3, -1

(B) $-3, 1$

(C) 3, 1

(D) $-3, -1$

Answer: (A) 3, -1 — $x = [2 \pm \sqrt{(4+12)}]/2 = [2 \pm 4]/2 \Rightarrow x = 3$ or $x = -1$.

Q12. The speed of a boat problem 'a boat takes 1 hour more to go 24 km upstream than to return downstream' typically leads to which type of equation in the boat's speed x ?

(A) Linear equation

(B) Quadratic equation

(C) Pair of linear equations

(D) Cubic equation

Answer: (B) Quadratic equation — Setting up time = distance/speed for both directions and equating the difference to 1 hour gives a quadratic equation in x after cross-multiplication.

Q13. If one root of $2x^2 + px - 15 = 0$ is -5 , the value of p is:

(A) 7

(B) -7

(C) 5

(D) -5

Answer: (A) 7 — Substituting $x = -5$: $2(25) + p(-5) - 15 = 0 \Rightarrow 50 - 5p - 15 = 0 \Rightarrow p = 7$.

Q14. The equation $(x+2)^2 = 2(x-3)$ reduces to which standard form?

(A) $x^2 + 2x + 10 = 0$

(B) $x^2 - 2x + 10 = 0$

(C) $x^2 + 4x + 4 = 0$

(D) $x^2 + 2x - 10 = 0$

Answer: (A) $x^2 + 2x + 10 = 0$ — Expanding: $x^2 + 4x + 4 = 2x - 6 \Rightarrow x^2 + 2x + 10 = 0$.

Q15. A quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$) has real roots if and only if:

(A) $b^2 - 4ac < 0$

(B) $b^2 - 4ac = 0$ only

(C) $b^2 - 4ac \geq 0$

(D) $b^2 + 4ac \geq 0$

Answer: (C) $b^2 - 4ac \geq 0$ — Real roots (equal or distinct) exist exactly when the discriminant is greater than or equal to zero.

4. Methodology — Steps Followed

26. Reviewed CBSE's official 2025-26 sample papers and marking schemes for Mathematics, Science and Social Science to confirm the current five-section (A–E) design, marks split, and the newly introduced Biology/Chemistry/Physics and History/Geography/Political-Science/Economics section divisions.
27. Cross-checked the pattern against secondary CBSE-preparation sources (Learn CBSE, Shaalaa, Vidyarohi, Axisa Academy, PW Live) to confirm consistency across the 2025-26 session.
28. Compiled a subject-wise question-type report (Section 1) covering Mathematics, Science, Social Science, English and Hindi, noting where the numeric A–E format applies and where a skill-based (reading/writing/literature) format applies instead.
29. Selected one representative NCERT chapter per subject (Chemical Reactions and Equations, Quadratic Equations, Nationalism in India, Nelson Mandela: Long Walk to Freedom, □□□□□□ □□ □□□□□□) and drafted structured AI prompts for every question type used in that subject's paper (Section 2).
30. Used the Mathematics and Science prompts to generate and manually verify two full 15-question MCQ banks with answer keys and explanations (Section 3), checking each for NCERT-syllabus accuracy and single-correct-option validity.
31. Compiled the problem statement, ownership/confidentiality notice, report, prompts and final MCQ banks into this single Word document for submission within the 24-hour window.

5. Final Output Summary

Deliverable	Status	Location in this document
Subject-wise question-type report (5 subjects)	Complete	Section 1
AI prompt sets, 1 chapter per subject (5 subjects)	Complete	Section 2
Sample MCQs — Science (15 Qs, chapter: Chemical Reactions & Equations)	Complete, with answer key & explanations	Section 3.1
Sample MCQs — Mathematics (15 Qs, chapter: Quadratic Equations)	Complete, with answer key & explanations	Section 3.2
Ownership / confidentiality note	Included on cover page and here	Cover page